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**Cross-View Association of Identical Parts Across
Non-Overlapping 3D Scans: Identifiability,
Shift Ambiguity and Evaluation**

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Contents

Contents	iii
List of Figures	v
List of Tables	vii
1 Introduction	1
1.1 The problem	1
1.2 Related work	3
1.3 Contributions	4
2 The data, the rig and the task	7
2.1 The dataset, annotations and evaluation metrics	7
2.2 Acquisition geometry	9
2.3 Detection	9
Bibliography	13

List of Figures

1.1	Dual-scanner system	2
1.2	The shift ambiguity (schematic)	2
2.1	Two views of one scene in the world frame	11
2.2	Detection	12

List of Tables

2.1	Profile families of the dataset	7
2.2	Quantities of the rig and of the acquisitions	10

Chapter 1

Introduction

1.1 The problem

Industrial handling is increasingly automated. Repetitive and heavy tasks such as lifting or sorting parts, once assigned to human operators, are now delegated to robots to satisfy the cycle time required by the production line.

For a robot to grasp an object, it is necessary to determine the object's pose, which dedicated sensors measure as a position and an orientation relative to the robot's own coordinate frame.

For a single compact piece, one 3D sensor is usually sufficient, as its viewpoint can cover the entire object. On the other hand, a very long part cannot be captured from a single viewpoint, making a single scanner insufficient. The layout therefore requires two sensors, one for each end. The part is gripped precisely at these ends, whose faces also determine the 6D pose of a straight extruded profile up to the symmetry of the cross-section.

The setup consists of two structured-light 3D scanners, positioned at either end of a conveyor carrying bundles of six-metre metal bars of identical profile.

Each device is mounted on a dedicated carriage at one end of the bundle, facing the end faces of the bars rather than their length. The two carriages are separated by several metres along the axis of the bar. The part of the bundle that each scanner covers is often narrower than the bundle itself, so a single view does not always contain every bar. Because the two sensors face opposite ends, their respective viewpoints neither overlap nor capture any common surface. Finally, a dual-arm manipulator picks up each bar by gripping it at both ends. Figure 1.1 illustrates this setup.

The task is therefore to determine which detection from one scanner and which from the other belong to the same physical object. In this specific case, data association cannot be resolved using conventional techniques due to the absence of typical geometric or visual features.

Absence of epipolar constraint. The lack of overlap between the two views means that the scanners never see the same surface.

Visual appearance invariance. Because the bars originate from the same extrusion batch, end faces belonging to the same profile family (i.e., bars sharing the same cross-section) are identical up to saw marks.

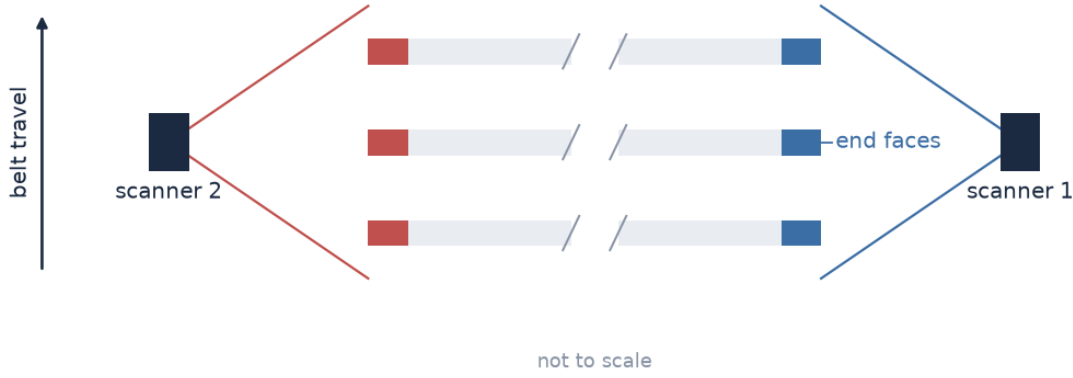


Figure 1.1: Dual-scanner system. Top view of the system. Each scanner is positioned at one end of the bundle and detects only the end faces of the bars facing it. In the figure, these faces are highlighted at both ends of each bar. The viewpoints of the two scanners do not share any surface and the bundle passes through both. Neither axis is to scale.

Relative position invariance. The bars are packed edge to edge at an almost uniform pitch, so the scanner detects a sequence of end faces that remains visually invariant even if the bundle shifts by one position.

Conveyor motion between acquisitions. Because the conveyor belt advances between the two acquisitions, the two scan windows can cut the bundle at different bars, in which case an association that pairs the detections in their order is shifted by one or more bars.

This shift ambiguity is illustrated in fig. 1.2. In each view, a different end bar remains excluded because each view often covers less than the whole bundle and the bundle moved between the two acquisitions. A correct association should record two unpaired detections, whereas a complete sequential association would lead to a systematic error.

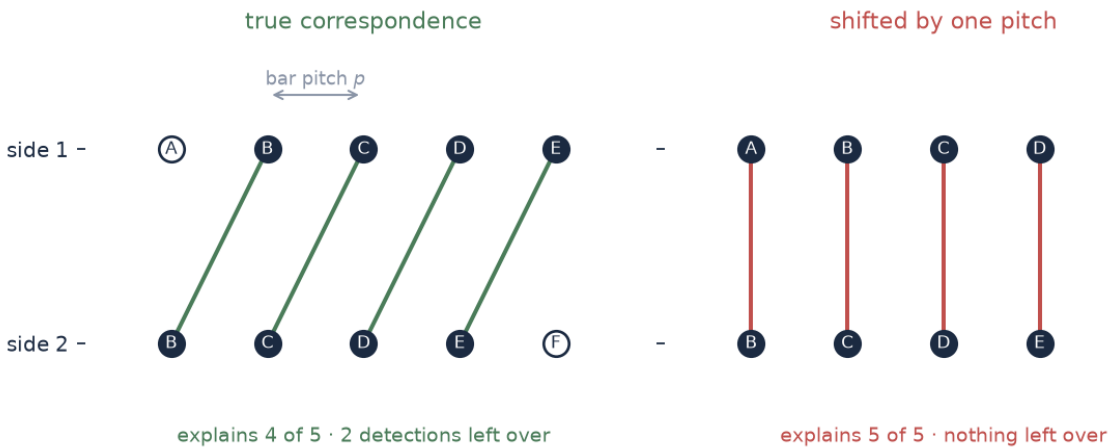


Figure 1.2: The shift ambiguity (schematic). Each scanner covers a different part of the bundle, so a different bar is truncated in each acquisition. A correspondence shifted by one pitch would incorrectly pair all detections, whereas the correct correspondence leaves the two truncated detections unpaired. Letters identify the physical bar and hollow circles mark the unpaired detections.

This condition is not an edge case because in many scenes of the dataset some bars are

scanned at only one end. The objective is to determine what portion of the correspondence is actually recoverable and to define how unpaired detections are handled.

The scans used in this thesis were provided by beanTech, the company that developed the vision software of the rig for the manufacturer that operates it. At the time of this work the rig was still being commissioned and only a limited number of scenes was available. This setting favours methods that require no training data, such as the classical detector and the pairing methods studied here, since the detector needs only the drawing of the bar cross-section.

1.2 Related work

To the best of our knowledge, no method in the literature addresses this specific case. Cross-view association is a widely studied problem, but existing approaches rely on assumptions that do not hold in this setting, such as the presence of a distinctive visual appearance or the geometric overlap of fields of view. Four main lines of research can nevertheless be identified, each resting on assumptions whose absence in this setting has specific consequences.

Geometric correspondence and assignment. The first line of research assumes the presence of surfaces observed in common. Methods based on stereo matching and multi-view correspondence, starting from dynamic programming along scanlines Ohta et al. 1985 [9], function precisely because the same surface appears in multiple images. The same principle applies to systems that associate identical instances detected by different cameras using geometry alone. These systems calculate the epipolar distance for each pair of detections, model it as a graph cost and recover the groups by clustering Sun et al. 2025 [13]. In the absence of visual overlap, however, the epipolar constraint no longer applies. The task thus reduces to a pure assignment problem, in which the cost matrix must be derived directly from the scene layout. The Hungarian algorithm solves minimum-cost bipartite matching Kuhn 1955 [4] without imposing order constraints on correspondences. An assignment can also be constrained to preserve the order of the elements, which is an explicit modelling assumption and does not follow from epipolar geometry. The monotonic alignment compared in this thesis (??) imposes such a constraint on the order of the bars.

Discriminative appearance. The second line of research leverages visual appearance to discriminate one object from another. Benchmarks for cross-camera instance association, such as MessyTable Cai et al. 2020 [1], are designed for objects showing appearance differences between instances, even minimal ones. According to their analysis, reliable association requires appearance, context and geometry together. The application most closely related to the case under consideration is found in the timber industry, where images of the same log end face taken at different times are matched as a biometric fingerprint, either with texture features that describe the annual ring pattern Schraml et al. 2015 [11] or with features learned by a deep network Xie et al. 2025 [15]. This approach is effective because each end face presents a unique surface, unlike a metallic extrusion whose cross-section is geometrically and visually identical to adjacent ones.

Identifiability and observability. The third line of research concerns the information content of the data themselves. In the absence of overlap and characterising visual attributes, the task reduces to an unlabelled sensing problem Unnikrishnan et al. 2018 [14]: observations are known, but the correspondence matrix is unknown. The fundamental question lies in determining what can actually be inferred from them. The literature addresses the topic along two directions: defining which configurations are reconstructible and formulating a converse that bounds what no algorithm can recover. The same critical issue emerges

in self-calibration for system parameter estimation, where it is addressed by studying the observability of the information matrix Maye et al. 2016 [7] and identifying unobservable directions in the parameter space Nobre et al. 2019 [8].

Association performance evaluation. The fourth line of research concerns association evaluation, which requires metrics distinct from those used for pure detection quality. Several studies on multi-target multi-camera systems highlight that single-frame metrics do not provide information on identity consistency across space and time. For this reason, identity-focused precision and recall metrics were introduced Ristani et al. 2016 [10], followed later by the HOTA metric, which prevents an aggregate index from masking the source of error between detection and data association Luiten et al. 2021 [6]. The BOP benchmark applies an analogous principle to 6D pose estimation, defining correctness in relation to the symmetry group of the object rather than relative to a canonical pose Hodaň et al. 2020 [3]. Finally, concurrent work shows that pairwise ranking metrics and one-to-one assignment metrics can disagree, which means that a model may achieve a higher score in ranking metrics without a corresponding improvement in its actual assignment capability Shelukhan et al. 2026 [12].

The gap addressed by this work lies between identifiability and evaluation. For an acquisition setup without visual overlap that observes identical elements at a regular spacing, it has not yet been formalised which part of the matching problem can be resolved in principle and which part is intrinsically undetermined. Consequently, reading an overall accuracy metric requires separating correctable algorithmic errors from structural limitations of the physical setup.

1.3 Contributions

This thesis determines what the data of two non-overlapping views can recover. The geometry of the rig is recovered except for the absolute position of the scanners, whereas when the bars are evenly spaced the geometry of the scene determines the correspondence between the detections only up to a shift by a whole number of bars. The pairing methods are then evaluated on the basis of this distinction rather than through a single aggregate accuracy figure. The contributions of this work are the following.

1. **The mechanism of the shift ambiguity.** Besides the true association, the data are compatible with a family of associations shifted by a whole number of bars along the bundle. When the bars are evenly spaced, a shifted association has practically the same geometric cost as the true one, because the translation estimated for the scene changes by the same shift and the distances between paired faces stay the same. The two associations differ only in the number of detections left unpaired, which is lower for the true association when both views contain the same bars. For this reason the ambiguity depends on whether the scan window truncates the two views differently and not on how far the bundle moves between the two acquisitions (?????).
2. **Confirmation on real data.** The errors of the most accurate pairing method on the real scenes follow this mechanism. Every scene on which the method fails completely is shifted as a whole and contains at least one detection without an annotated partner. In each of these scenes the wrong association also has a lower cost than the annotated one under the objective that the method minimises, so these errors depend on the information contained in the detections rather than on the search for the minimum (??).

3. **Identifiability of the rig geometry.** Prior knowledge of the nominal bar length determines the separation between the two carriages but not their absolute position, which the scene content does not constrain. Alternating calibration and assignment recovers the separation and removes an error of about one metre along the bar axis (????). The length check of the original system cannot detect a shifted association, because pairing the ends of two adjacent bars changes the reconstructed length by far less than the tolerance of the check (??).
4. **The hybrid method.** The pairing method proposed in this thesis applies the order constraint of the monotonic alignment within each stacking level and admits a bar crossing its neighbours only when the detections support it (??). With both of the detectors compared in this thesis it is the most accurate of the methods tested, although on this dataset its advantage over pairing the detections by their order alone is not statistically significant (????).
5. **Evaluation of the pairing methods.** The counting metric checks only how many pairs are assigned and therefore does not distinguish a correct association from a shifted or swapped one. This thesis relies instead on a correspondence annotation linked to the detections by position, which makes it possible to score the output of any detector against the same ground truth (section 2.1 and ??).
6. **Development and audit of a synthetic twin.** The synthetic twin, developed to simulate scene acquisitions, replicates detection counts and detector accuracy but does not model how far the bar ends stagger along the bar axis. This restricts the scope of any validation conducted on it to the transverse plane alone (??).

Chapter 2

The data, the rig and the task

Apart from the controlled experiments on the synthetic twin (??), all results in this thesis come from real acquisitions. Every comparison is between pairing methods that operate on the same detections and return cross-view pairs. A detection is an end face identified in a single view, whereas a pairing method is the procedure that takes the detections from both views as input and returns a set of pairs connecting them.

2.1 The dataset, annotations and evaluation metrics

A scene consists of a bundle of bars acquired once from each end, yielding two point clouds. The dataset comprises 108 scenes and 216 clouds, classified by profile family (table 2.1), with each scene associated with a manually assigned scenario category.

Table 2.1: Profile families of the dataset. For each family, the shape of its cross-section as given by the CAD drawing, the number of scenes and the number of annotated faces. The L-profile appears twice because it was acquired in two separate campaigns, so the five families share four distinct cross-sections. The five families total 108 scenes and 935 faces.

profile family	cross-section	scenes	annotated faces
complex profile	complex	4	15
heavy profile	rectangular, 80×95 mm	11	68
L-profile	asymmetric L-shape	40	274
L-profile, second acquisition	asymmetric L-shape	24	223
square profile	square, 40×40 mm	29	355

The second column indicates the cross-section shape, obtained from the CAD drawing of the profile. Although the dataset consists of five families, there are only four distinct cross-sections because the L-profile appears twice due to two separate acquisition campaigns.

The categories describe the arrangement of the bars in the bundle: single (56 scenes), overlapping (22), standing (15), crossed (9) and tilted (6). Across all scenes, a single side features a median of 4 faces and a maximum of 29.

Truncation, where a bar is visible to only one scanner, affects 30 of the 108 scenes and leaves 49 of the 935 faces without a counterpart in the other view. Consequently, the assumption that every bar is visible to both scanners fails in roughly one out of four scenes.

Two types of annotations have been defined.

The first is a count annotation, indicating how many end faces are visible for each side. It was made first, by hand, on images generated from the point clouds and independently of any detector.

The second is a correspondence annotation that, for each scene, provides a list of claims: inter-view pairs, each equipped with an anchor point on both faces, taken from the centroid of the face detected by the candidate-pool detector and confirmed by the annotator. In total, this annotation contains 407 claims, with at least one in every scene. The anchors allow each claim to be associated with the detections of the detector being evaluated, verifying whether the detector successfully produced that pair. A claim is resolved for a detector when both of its anchors can be linked to a detection of that detector. Where a scene needs it, the annotation also carries a note recording, for example, which profile is truncated in one view or not confirmed by the detector.

The two annotations support different metrics. A counting metric checks only how many pairs a method produces, so the criterion is satisfied whenever the number of pairs is correct, even if all pairs are incorrect. For example, if two true pairs are swapped, the method still produces two pairs and the counting metric counts them as two true positives. The same happens when one view truncates a bar that the other contains, in which case an association shifted by one position along the row keeps the annotated pair count. Only the correspondence annotation indicates which pairs are correct and can therefore detect a swap. Work on multi-camera tracking faced a similar limitation and resolved it with identity-based precision and recall metrics Ristani et al. 2016 [10].

Four metrics are evaluated:

- **Counting precision, recall and F1:** compare the number of faces or pairs produced by a method against the count annotation, both scene-by-scene and aggregated. A true positive is defined as the smaller of the two numbers, calculated per side for faces and per scene for pairs.
- **Correspondence recall:** the proportion of claims correctly reproduced by a method (i.e., with correct bar identity), evaluated over the set of resolved claims. Restricting the score to resolved claims prevents a missing detection from being counted as a pairing error, following the distinction formalised by HOTA for tracking Luiten et al. 2021 [6].
- **IPAA-1.0** (Image-Pair Association Accuracy at 1.0): the proportion of scenes correctly resolved in their entirety (referred to as IPAA-100 in MessyTable Cai et al. 2020 [1]). While aggregated recall gives greater weight to scenes with many associations, IPAA-1.0 weighs every scene equally. Furthermore, it aligns most closely with a line operator’s experience, where an entire bundle is processed as a single unit.
- **Inversions:** the count of pairs within the same row that lie side-by-side at the same height in both views but exhibit an inverted Y-order between them. These capture the swaps that counting metrics cannot detect by design.

In this thesis, all confidence intervals are calculated by resampling bundles rather than pairs. Pairs within the same scene share the same bundle, transport estimate and assignment, so they are not independent samples and an error tends to affect the whole scene rather than individual pairs. Some scenes are also repeat captures of the same stationary bundle, whose point clouds coincide within 2 mm on almost every point. These scenes are grouped and resampled together, so the 108 scenes of the dataset form 96 independent units, called bundles

in the rest of the thesis. For correspondence metrics all of them are resampled, since each scene contains at least one claim.

All results reported here are based on the candidate-pool detector (section 2.3). The anchors were defined from its output, so every claim corresponds to one of its detections and all 407 claims can be scored.

The alternative detector, based on two-pass suppression (section 2.3), resolves 382 of the 407 claims and re-associates faces with a median error of 0.96 mm, compared to 0.00 mm for the candidate-pool detector. The comparison between the two detectors therefore measures how robust each method is to a change of detector rather than which detector is more accurate.

2.2 Acquisition geometry

Two scanners are placed at opposite ends of a conveyor belt, each mounted on its own linear carriage. The global reference frame (in millimetres) is defined relative to the robot: the X-axis aligns with the bars and carriage motion, the Y-axis indicates the conveyor feed direction and the Z-axis represents height. Calibration relates each scanner’s local frame to the global frame via a rigid transformation $p_{\text{world}} = R p_{\text{scanner}} + t$. After calibration, the point clouds are orthographically projected onto the YZ plane (feed and height), which lies nearly frontal to the end faces.

Two calibrations are available. The offline calibration, defined by a single rotation and translation knot per scanner, yields a constant rigid transformation valid exclusively at the carriage position where knots were acquired. By contrast, the original system calibration uses ten knots per scanner across roughly 2.1 m of travel, interpolating between them with a centripetal Catmull-Rom spline for translation and spherical cubic interpolation (SQUAD) for rotation, with values clamped at the boundaries. Applying the original calibration would require the carriage position of every scan, which is absent from the dataset, so the constant offline calibration was used instead. Consequently, accuracy is limited to the single acquired position, introducing an error along the X coordinate, the axis of carriage motion, that is subsequently measured and compensated (??). Figure 2.1 shows the two clouds of one scene in the world frame.

Several quantities in table 2.2, along with certain detection constants, originate from the source code of the original system installed on the machine.

2.3 Detection

The points that survive the radial filter are projected onto the YZ plane and a look-up table records which 3D points correspond to each pixel, enabling a 2D match to retrieve its 3D support. The profile cross-section is rasterised from the CAD drawing (DXF file) at 2.5 px/mm and slid over the image at 2° intervals across the full rotation. Every local cross-correlation maximum reaching the lost-face threshold of 0.2 constitutes a candidate (the threshold is defined relative to the template’s self-correlation peak).

The two detectors used in this work share this initial stage but differ in the criterion used to confirm candidates.

The two-pass suppression detector decides on the 2D image: it confirms a candidate when the peak reaches the match threshold of 0.4 and the mask overlap exceeds 0.65, suppresses

Table 2.2: Quantities of the rig and of the acquisitions. For each quantity, its value and how it was obtained. Lengths are in millimetres in the world frame, where X runs along the bars, Y along the conveyor feed and Z in height.

Quantity	Value	How it was obtained
Standoff	656 mm, interquartile range (IQR) [641, 669] mm	Median radial distance between the scanner origin and the retained points across 216 clouds.
Angle of incidence on the end face	22.0°	Extracted from the production calibration rotation matrices. Stable within $\pm 0.35^\circ$ over roughly 2.1 m of carriage travel.
Angular envelope	$[-19.3^\circ, +17.4^\circ]$, union of both sides	Field of view width of each scanner along its tangential direction, within its local frame. The lower limit repeats within 0.06° across 104 of 108 scenes on both sides, indicating a hardware limitation.
Scan window	median 258 mm, maximum 414 mm	Calculated between the 0.5th and 99.5th percentiles of points retained after the 850 mm radial filter. Prior to filtering, the median extent is 479 mm. The effective window depends jointly on this radial filter and the device’s angular span.
Bar pitch	[38.6, 118.4] mm	Range of median distances between adjacent bars, calculated for each of the five families on the correspondence anchors.
Cross-view transport	median $ \Delta Y $ 32.1 mm	Calculated per scene as the absolute value of the median ΔY between the anchor points of the two views. The reported value represents the median of these values across all 108 scenes.
Nominal bar length	6005 mm	Value set by the operator in the original system, independent of the profile family.
Distance between carriages	5403.3 mm	Estimated via self-calibration (??) rather than direct measurement. It differs by 1.3 mm from the documented separation of 2701 mm per side (5402 mm total).
Effective resolution	2.5 px/mm	Product of the rendering scale (10 px/mm) and the downscaling factor (0.25).

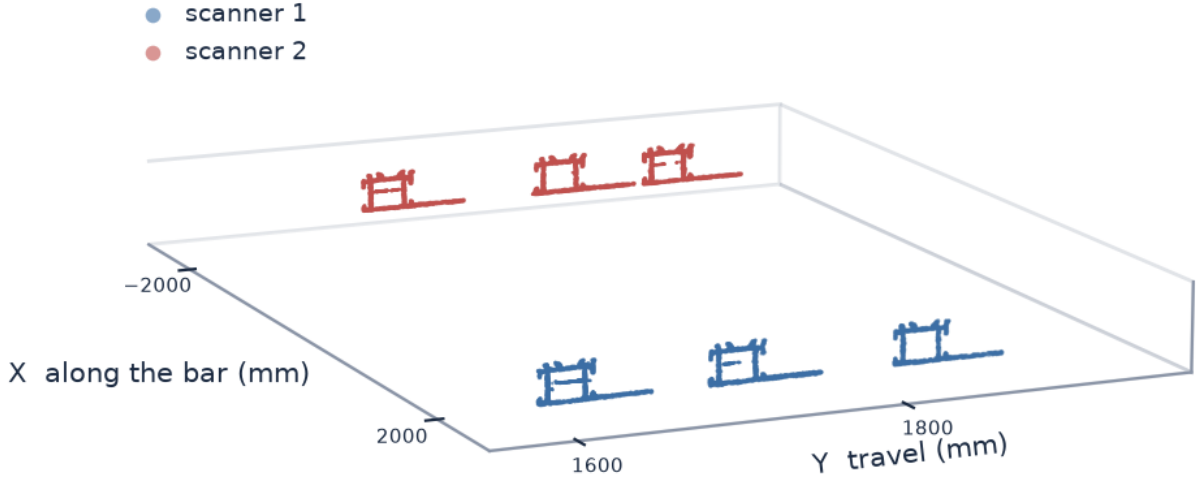


Figure 2.1: Two views of one scene in the world frame. The two point clouds of a single scene in the global reference frame. The radial filter, applied before calibration, discards any point beyond 850 mm from its scanner, which isolates the end faces from the machine and the conveyor belt and leaves 15,037 of the 298,061 points on the first side. Nothing is ever acquired between the two clouds, so the bar is never fully scanned. The scene belongs to the L-profile family.

duplicates at an intersection over union (IoU) of 0.2, clears the confirmed footprints from the image and repeats the sweep so that a hidden face behind a stronger neighbour can emerge in the second pass.

The candidate-pool detector instead decides based on the 3D points for which candidates compete. It retains every candidate above 0.2 without angle limits or bounding box overlap suppression, assigning to each candidate the 3D points within its footprint at its depth. Finally, a single global selection picks the subset explaining the most points against a fixed per-instance cost, with the constraint that no two selected candidates claim the same point. The candidate-pool detector is the one used in the remainder of this work, while two-pass suppression was retained to repeat the comparison between pairing methods on a different underlying detector. Figure 2.2 shows how the two detectors interpret the same image.

For each detected face, both methods return the 3D footprint points by masking the crop with the template shape and tracing back through the look-up table. Extracting the centroid and normal of the fitted plane via total least squares yields inputs for the pairing gates, combined with the matching angle retained from the 2° quantised sweep.

In this detection phase, the resolution, the 2° step size and the correlation thresholds come from the original system installed with the machine. After identifying a face and retrieving its 3D points, that system executed an extra step: it re-projected the face onto its own plane, estimated using RANSAC Fischler et al. 1981 [2], and repeated the angle search on that view with a finer step. It thus obtained a refined angle and a planar reference for each face. The detection pipeline used in this thesis does not perform this step. A more precise angle would tighten the consistency checks between the ends of the same bar without helping to decide which end belongs to which bar, since identical parallel bars share the same angle.

Across all 108 scenes, the candidate-pool detector achieves a precision of 0.986, a recall of 0.955 and an F1 score of 0.970 against per-side counts. A subsequent recovery stage increases recall to 0.967 and F1 to 0.976. Operating on unpaired faces, this stage estimates

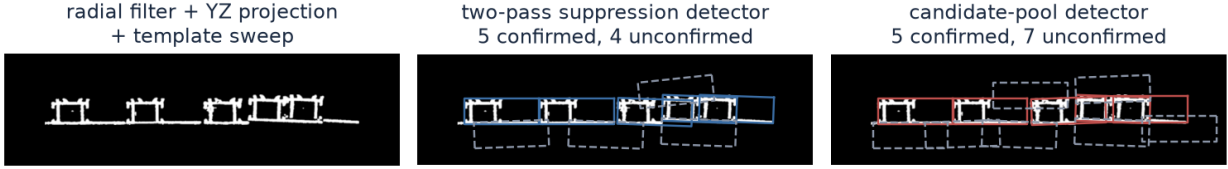


Figure 2.2: Detection. On the left, the projection on which both detectors operate after the radial filter and the template sweep. In the centre and on the right, the interpretation of the same image by each detector: confirmed detections are enclosed in coloured bounding boxes, while unconfirmed candidates appear in dashed grey boxes. In this case, both confirm the same five faces, but differ in the amount of residual unconfirmed evidence: four candidates versus seven.

their position in the opposing view using the cross-view translation, searches for a local match and accepts it only if the 3D footprint exhibits valid end-face characteristics. Approximately 20% of attempts are accepted.

As a baseline for the classical detector, a small convolutional network was trained on the same annotations and with the same code. It predicts face centres as peaks of a Gaussian heatmap, with a second head recovering their offset within the output cell. This formulation follows the centre-point detection proposed in Zhou et al. 2019 [16], whose Gaussian target and loss function derive from Law et al. 2018 [5]. Detection labels give only face centres, so a detector predicting bounding box extent and orientation would lack data to learn those parameters. Direct centre prediction instead keeps both model branches focused on per-side counts and the position of each face.

The results of the learned model depend on how the data is split. In the scene-disjoint split, where the training and test scenes differ but contain the same cross-sections, the model achieves a test F1 of 0.902. In the family-disjoint split, the test family is never seen during training and the two L-profile acquisitions are held out together, yielding an F1 of 0.425. The drop from 0.902 to 0.425 shows that the model recognises previously seen sections well, but fails to generalise to novel geometry. Both values are averages across two training runs, in each of which the model sees only a subset of the annotations, and should be compared with the classical detector’s 0.970. A paired bootstrap over bundles on one of the two runs gives, in both splits, a confidence interval for the difference that lies entirely below zero, so the learned model is significantly less accurate than the classical detector.

The discrepancy between the two splits shows that the model learns the four cross-sections present in the dataset, but does not develop a general end-face detector. Given a DXF drawing, the classical detector requires no training data and therefore always operates in the family-disjoint regime. The price is inference time, which is 2.53 s per scene side against 18 to 24 ms for the learned model. The two F1 values are lower bounds, because training was stopped before the test F1 had converged under a CPU-only schedule. The qualitative conclusion does not depend on them, since the model does not generalise to an unseen cross-section.

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